Supplementary Material: Humans Are More Diverse: Frontier LLMs Show Extreme Policies in Idealised AI Development Races

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ABSTRACT

This document holds the material the main paper points to but has no room to carry: the full breakdown of every task-validity probe, the replication check against the human benchmark, a claim-by-claim audit of the citations, the persona and measured-risk comparison, the predictive analysis behind the reported drivers, and the complete descriptive baselines for two-player and N-player races. Nothing here is needed to follow the main argument. Everything here comes from the same runs, under the same admission rules, and is reported at the same evidence level.

KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, task validity

HOW TO READ THIS DOCUMENT

Each section stands on its own and is named for what it measures. Section letters restart at A because this is supplementary matter.

- **Task-validity probe battery.** Every probe, its wording family, and the per-model pass rate behind the summary in the main paper.
- **Human-benchmark replication check.** Whether the reference human study's reported pattern reproduces on the released participant data under our own analysis code.
- **Claim-scoped citation audit.** For each cited work, the claim it is used to support and the claim it does not support.
- **Risk preference and assigned personas.** The persona families, and how a prompted persona compares with measured human risk preference.
- **Predictive structure of unsafe choices.** The feature analysis behind the drivers reported in the main paper.
- **Two-player descriptive baseline, full scope.** Every checkpoint and condition, including those the main paper does not plot.
- **N-player self-play scope.** Group sizes three to five.
- **Trajectory embedding recoloured by outcome.** The same embedding as the main paper, coloured by unsafe rate instead of population.
- **Unsafe rate by persona band and relative position.** The cell-level numbers behind the grouped-bar figure in the main paper.

A EXTENDED RESULTS AND ROBUSTNESS CHECKS

This appendix collects the material referenced from the results section in the main paper as supporting rather than extending the main claims: the full task-validity probe breakdown (§A.1), the human-benchmark replication check (§A.4), the persona comparison (§A.6), the predictive analysis (§A.7), and the full two-player and N-player descriptive results (§A.8, §A.9).

A.1 Task-validity probe battery: full breakdown

Table 1 reports the full breakdown behind the frontier endpoint admission in the main paper. The confirmatory audit used the the two exact routes listed in Table 2 under the frozen protocol v5: 20 atomic probes spanning rule recall, one-stage payoffs, state reconstruction, state transition, terminal scoring, and expected payoff, repeated three times per route for 120 retained outputs. Every response was schema-valid and no transport retry was needed. Numeric probes used a fixed structured-output contract; categorical probes explicitly listed the allowed answer set.

Table 1: Task-validity probe accuracy by category for the two admitted frontier routes, protocol v5 (n = 60 retained probe outputs per route).

Probe category	Gemini 3 Flash	Claude Sonnet 5
Rule recall	100.0%	100.0%
One-stage payoff lookup	100.0%	100.0%
State reconstruction	93.3%	86.7%
State transition	100.0%	100.0%
Terminal scoring	100.0%	80.0%
Expected-payoff calculation	50.0%	0.0%
Overall semantic accuracy	*93.3%*	*81.7%*
Schema-valid responses	*100.0%*	*100.0%*

The largest errors were in the optional payoff check: three of six Gemini responses and none of the Claude responses were correct. We retain this domain as a diagnostic. Live gameplay asks for an action, whereas the state and terminal domains determine whether the action prompt is mechanically interpretable. The admission task therefore records both the required gate and the arithmetic limitation instead of silently dropping the endpoint.

A.2 Frontier endpoint registry and admission outcomes

Endpoint availability was checked against the active Kaggle Benchmark identity and recorded using the exact route string. One repetition of the 20-probe bank was used as a diagnostic smoke for every advertised route; the two routes that passed the v5 gameplay thresholds then received the three-repetition confirmatory admission. A route that was unavailable or failed transport was retained as failed infrastructure evidence, never replaced by another endpoint.

The required gameplay gate was overall accuracy at least 80%, state reconstruction accuracy at least 75%, and terminal-scoring accuracy at least 75%. Expected-payoff accuracy was reported separately because it is a closed-form arithmetic diagnostic rather than

Table 2: Per-endpoint frontier admission audit. Smoke results are diagnostic; only routes with a completed v5 artifact and passing gameplay thresholds enter the new gameplay run.

Exact route	Overall	State	Terminal	Status
google/gemini-3-flash-preview	93.3%	93.3%	100%	admitted
anthropic/claude-sonnet-5@default	81.7%	86.7%	80%	admitted
google/gemini-3.1-flash-lite-preview	75%	60%	80%	diagnostic
openai/gpt-5.4-nano-2026-03-17	50%	20%	60%	diagnostic
deepseek-ai/deepseek-r1-0528	–	–	–	transport failed
ibm/granite-4.0-h-small	–	–	–	transport failed
openai/gpt-oss-120b	–	–	–	transport failed
qwen/qwen3-next-80b-a3b-instruct	–	–	–	transport failed

an action-task requirement. The active identity exposed no second authorised account, so no multi-account pooling is claimed.

A.3 Frontier neutral baseline

The new neutral gameplay baseline used protocol v3, the same mechanism and prompt for both admitted routes, ten repetitions per risk level, and 30 races per route. Each route produced 558 structured decisions with zero parse failures. The entries in Table 3 are decision-weighted Unsafe rates; they are not pooled across routes and do not estimate an evolutionary equilibrium.

Table 3: Neutral two-player frontier baseline under the admitted routes. Each route has 30 races, 558 decisions, and zero parse failures.

Exact route	$r = .1$	$r = .6$	$r = .9$	Races
google/gemini-3-flash-preview	98.9%	74.2%	59.1%	30
anthropic/claude-sonnet-5@default	89.2%	46.2%	37.6%	30

The Gemini manifest records that the hosted SDK stripped the requested sampling seed. The Claude manifest records that the seed was forwarded but provider-side application was not confirmed. Temperature was requested as 0.7 for both routes but was not forwarded by this SDK path. These facts bound the result to the exact hosted route and software contract.

To make the comparison boundary explicit, Figure 1 separates the risk response, the route-specific departure from the strong-selection reconstruction, the nearest-rule diagnostic, and its mismatch rate. This is an analysis artifact rather than an equilibrium test: the frontier routes generate prompted self-play trajectories, whereas EGT samples stationary population composition.

The invasion view in Figure 2 exposes the selection topology behind the three theoretical endpoints. Its directed edges are computed by the archive-compatible EGTtools implementation. The node areas use the finite-mutation main-reference chain because the small-mutation limit can contain multiple absorbing classes at extreme payoff scales. This is a diagnostic decomposition, not a claim that the frontier routes sample the same population process.

A.4 Human-benchmark replication check

Before comparing model-generated behaviour with the human benchmark of Domingos and Han [4] in the two-player strategic diversity results in the main paper, we first validated our reconstruction of that benchmark by refitting its published dynamic specification on the same public de-identified data. The reconstructed estimation sample contains 2,888 round-level observations, 172 race-pair clusters, and 338 participants. Table 4 compares the refit coefficients against the values reported in the original study.

Table 4: Replication of Domingos and Han [4]'s dynamic specification on the public de-identified human data. "Reconstructed" is our refit; "Original" is the coefficient reported in the source study.

Term	Reconstructed	Original
Opponent's previous action	0.606	0.607
Player's progress gap	−0.295	−0.296
First-round action	0.217	0.217
Player's own previous action	−0.195	−0.193

A.5 Claim-scoped citation audit

Table 5 records both the supported use of each cited study and the nearest excluded overclaim. These evidence boundaries do not turn the diagnostic pilots in this paper into confirmatory results.

The reconstruction reproduces the original coefficients almost exactly (largest discrepancy 0.002), which is the basis for treating the human-side comparisons in the two-player strategic diversity results in the main paper and the persona and measured risk results in the main paper as resting on a correctly processed benchmark rather than on a reimplementation error.

A.6 Risk preference and assigned personas

We finally distinguish three constructs that should not be conflated: the human participants' pre-game elicited risk preference, the mechanism's assigned private-risk treatment, and the narrative risk persona supplied to an LLM in its prompt. Human participants completed an incentivised Eckel–Grossman gamble elicitation (0 = most risk-averse, 5 = least risk-averse) before playing the two-player race. This elicited measure has essentially no direct association with subsequent UNSAFE rate,

$$\text{slope} = -0.004, \qquad r = -0.015, \qquad p = 0.79, \qquad n = 341,$$

with the difference in mean UNSAFE play between the most and least risk-accepting human groups (approximately five percentage points) lying within the corresponding uncertainty interval. This null result independently reproduces Domingos and Han's own preregistered finding that elicited risk preference did not significantly predict UNSAFE choice [4], and stands in contrast to the tournament literature's general finding that human risk-taking responds to competitive standing rather than remaining a fixed trait [5, 7, 8]: in this task, the round-by-round interaction captured in the two-player strategic diversity results in the main paper and the multi-agent position results in the main paper is a stronger organising signal than static pre-game disposition. A weaker, second-order association does survive at the level of behavioural phenotype rather than mean rate: the same elicited measure differs across the raw-sequence archetypes of the two-player strategic diversity results

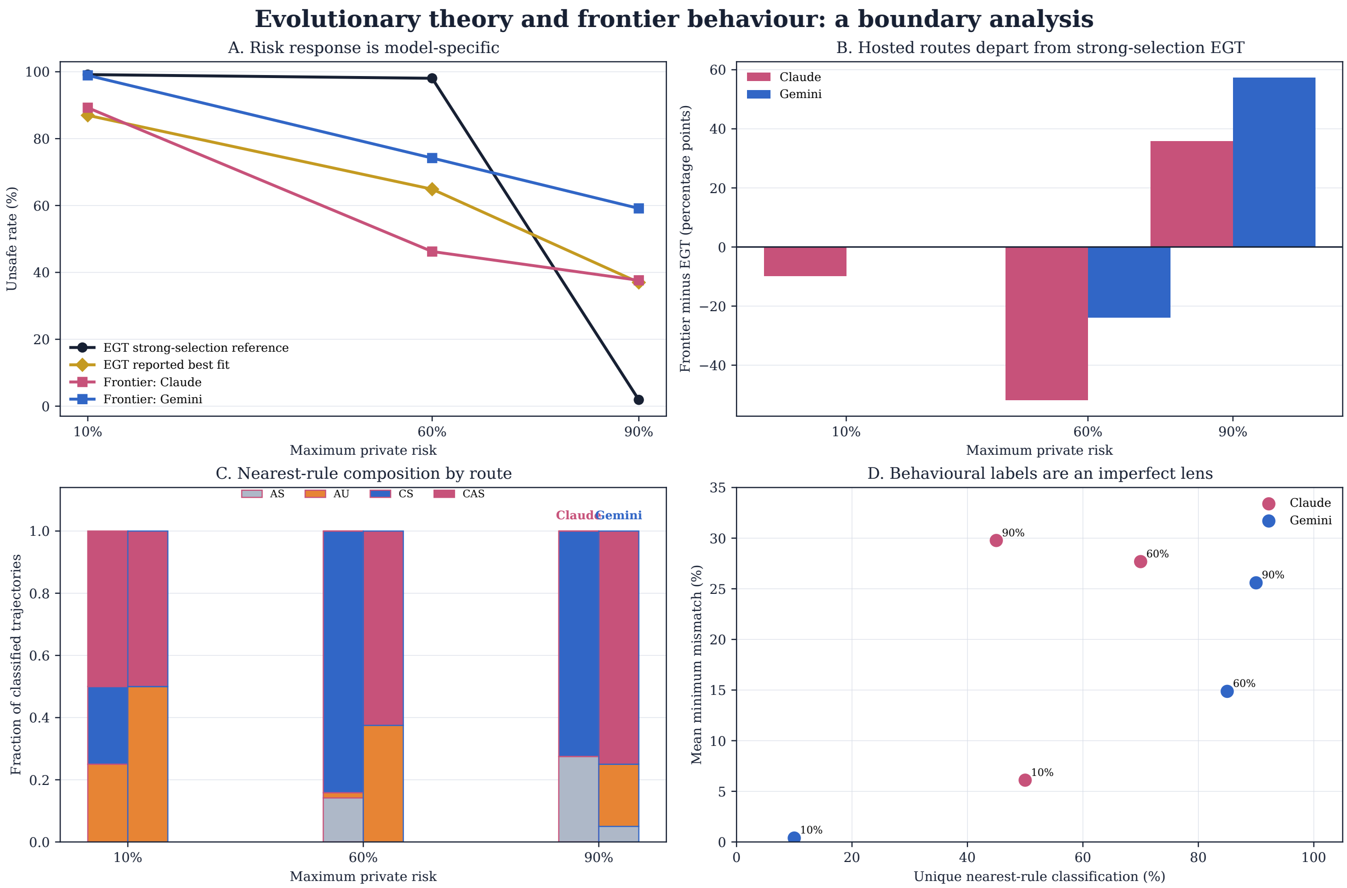

Figure 1: Boundary analysis for the reconstructed evolutionary model and the two admitted frontier routes. Panel A shows route-specific Unsafe responses alongside the two EGT parameter regimes. Panel B shows the deviation from the strong-selection EGT reference. Panel C gives the nearest-rule composition, and Panel D shows why those labels are only a diagnostic lens: uniqueness and minimum mismatch vary across routes and risk levels.

in the main paper (Kruskal– Wallis $H = 21.95$, $p = 0.001$, $n = 286$ clustered human players), so the elicitation may help stratify *which* dynamic pattern a participant falls into even though it does not predict *how much* UNSAFE they play overall.

Figure 3 (left) shows how little that null leaves. Human mean UNSAFE play stays inside a 49–62% band across the entire elicitation range, never approaching the low-UNSAFE models of the two-player strategic diversity results in the main paper (GPT-5-nano 17%, Claude Sonnet 5 22%, Claude Opus 5 33%) or the high-UNSAFE Gemini models (73–83%). Elicited risk preference does not move a participant far enough to resemble a different model's policy.

The right panel is the opposite (Table 6): every model with a six-level sweep climbs 50.5 to 98.3 points from R1 to R6, all but the two nano models monotonically, and those two dip by well under a point at R2 before rising. The label does not, however, move models by a common factor; GPT-5.6 Terra runs lowest at R1 and finishes within 1.3 points of the top, while GPT-5-nano starts fifth of seven and ends last last, so curve order is not preserved, and the same instruction is worth very different amounts to different models. Of the seven tested models, five have such a sweep; neither Gemini Flash-Lite checkpoint does. In logistic models that additionally control for the mechanism's actual private-risk treatment, the coefficients on the agent's own assigned risk label remain large for the three models where that controlled fit has been run, so the persona effect is not simply standing in for the mechanical risk treatment.

This contrast does not show that LLM agents possess a latent risk preference comparable to the human elicitation: the human variable is a measured pre-game disposition that turns out to be nearly unrelated to in-game behaviour, whereas the LLM variable is an explicit instruction embedded in the prompt. The more defensible reading is that an assigned risk persona functions as a high-salience behavioural instruction for these models, not as an analogue of a stable human risk disposition. Because the LLM estimates come from a separate risk-matrix experiment, available for five of the seven tested models and with the private-risk-controlled logistic fit run for only three of those five, this comparison remains exploratory and cross-experimental rather than a like-for-like measurement.

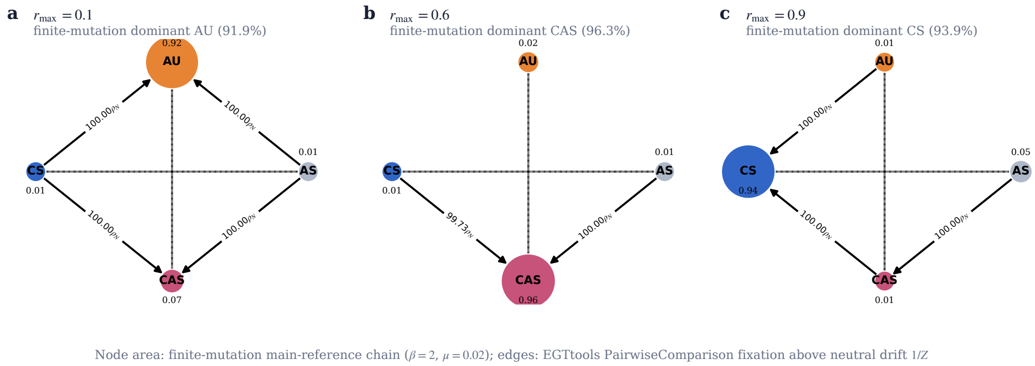

Figure 2: Risk-conditioned invasion topology for the reduced evolutionary game. Node area shows the finite-mutation main-reference chain share ($\beta = 2$, $\mu = 0.02$); directed edges show EGTtools `PairwiseComparison` fixation probabilities above neutral drift. The figure is a hybrid diagnostic: the edge topology is evaluated in the EGTtools small-mutation limit, while node areas come from the independently simulated finite-mutation chains. It should not be read as a latent-strategy diagram for the language models.

Table 5: Source-level audit for citations added to the prompt-sensitivity and behavioural-validity argument.

Source	Supported use	Excluded overclaim
Sclar et al. [12]	Meaning-preserving few-shot formatting changes can yield large, model-dependent performance spreads.	Not evidence for semantic paraphrase effects or a universal effect size.
Pezeshkpour and Hruschka [9]	Reordering multiple-choice answer options can change predictions.	The proposed uncertainty-plus-placement mechanism is not causally identified.
Wei et al. [13]	Order and response-token identity are distinct selection-bias targets.	Not proof that every task or model exhibits the same bias.
Salinas and Morstatter [11]	Controlled output-format and atomic whitespace variants can alter answers.	Not a claim that every added space changes behaviour.
Zheng et al. [15]	Across their factual-question setup, average persona benefit was absent or slightly negative and reliable persona selection remained difficult.	Not a universal null effect for persona-driven agent tasks.
Aher et al. [1]	Three of four behavioural findings were qualitatively reproduced, while the fourth showed a hyper-accuracy distortion.	Not evidence that LLM samples are interchangeable with human populations.
Akata et al. [2]	Prediction before action improved coordination and scores in repeated games.	Not a general claim that an observed action reveals a correct opponent belief.
Herr et al. [6]	Action-label order and payoff reassignment changed choices in two canonical two-player games.	Not evidence that every strategically equivalent transformation has the same effect.
Robinson and Burden [10]	Procedurally varied vignettes produced substantial contextual variability under one underlying Prisoner's Dilemma structure.	Not a repeated-game result and not an estimate for the present race.
Wu et al. [14]	Performance declined consistently across 11 tasks when default mappings were replaced by specified counterfactual alternatives.	Not proof that the tested models lacked all abstract reasoning ability.
del Rio-Chanona et al. [3]	LLM markets recovered broad feedback-condition dynamics but displayed less strategy heterogeneity than human participants.	Not evidence that LLM agents and human participants are exchangeable.

A.7 Predictive structure of UNSAFE choices

The preceding section shows that populations produce different action patterns. We now ask which recorded parts of the game best predict their next Unsafe choice. This is a study of association, not cause or internal reasoning. For each population, we fit a random forest, a model that combines many decision trees. It predicts decisions after round one from five values known before the choice: the player's previous action, the opponent's previous action, the progress gap, the assigned maximum private risk, and the round number. TreeSHAP then measures how much each value contributes to the fitted model's predictions. A larger SHAP share

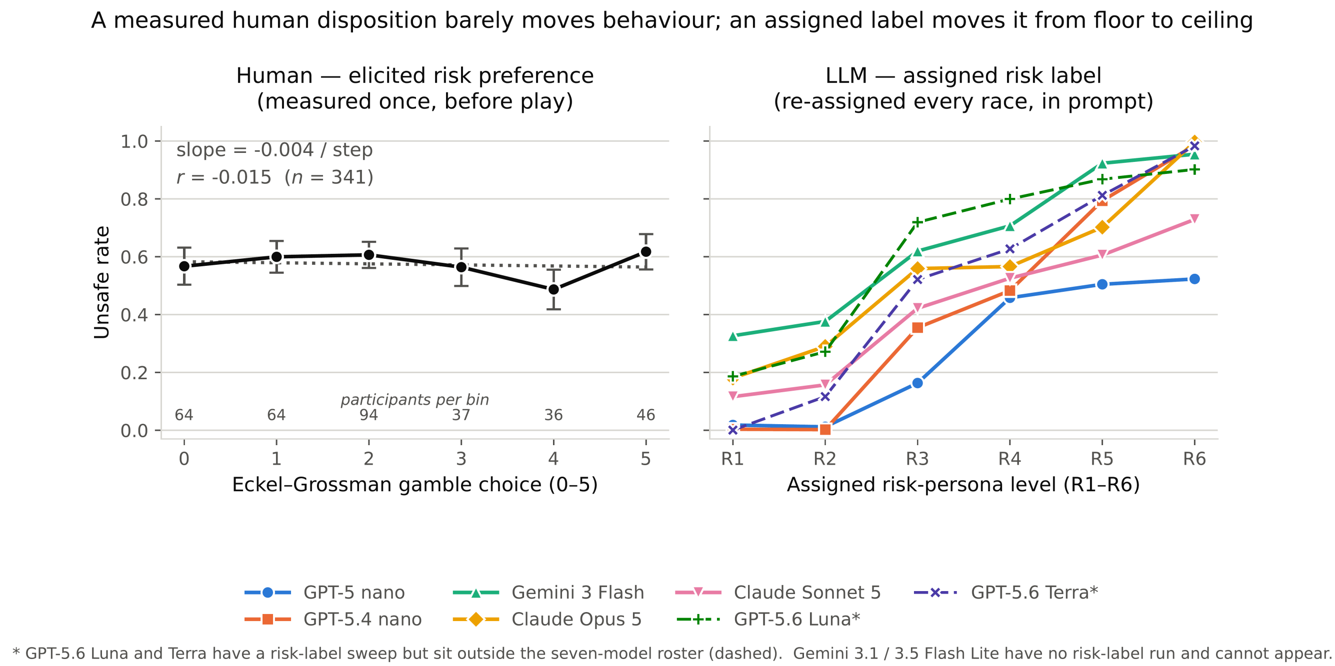

Figure 3: A measured human disposition against an assigned model label. Left: mean UNSAFE rate by elicited Eckel–Grossman gamble choice, with 95% confidence intervals and participants per bin; the dotted line is the trend across the six group means. Right: mean UNSAFE rate by assigned risk-persona level, for every model with a six-level sweep; GPT-5.6 Luna and Terra (dashed) sit outside the seven-model roster. The shared six-step axis is visual alignment only; the left is a disposition measured once before play, the right an instruction re-assigned every race.

Table 6: LLM risk-persona effect: change in UNSAFE rate from the lowest (R1) to the highest (R6) assigned risk label, and the per-level OLS slope, for every model with a six-level risk-label sweep ($n = 360$ players per level, $n = 330$ for Gemini-3-Flash). The persona coefficient (logistic model additionally controlling for the mechanism's actual private-risk treatment) has been fit only for the first three; "–" marks models for which only the descriptive delta and slope are available. Starred rows sit outside the seven-model roster.

Model	Δ UNSAFE (pp)	Per-level slope	Coefficient
GPT-5-nano	50.5	0.123	0.93
GPT-5.4-nano	98.2	0.212	1.47
Gemini-3-Flash	62.8	0.139	2.03
Claude Opus 5	81.6	0.152	–
Claude Sonnet 5	61.3	0.129	–
GPT-5.6 Luna*	71.5	0.156	–
GPT-5.6 Terra*	98.3	0.203	–

means that the predictor relied more on that feature. It does not mean that the feature caused the decision.

For human participants, the opponent's previous action accounts for 48% of total SHAP magnitude, far more than any other feature, while the player's own previous action contributes only 6%: a reciprocity-dominated profile that independently recovers the pattern found in the human logistic analysis. GPT-5-nano is close to the opposite profile: relative race position (progress gap) accounts for 44% of its predictive importance and opponent history for only 4%, and its apparently strong test AUC of 0.82 is misleading on its own, since balanced accuracy is exactly 0.50: the model plays so close to the SAFE floor that the classifier effectively fails to recover the rare UNSAFE class. Gemini-3- Flash and Gemini-3.1-Flash-Lite are primarily risk-treatment-driven (35% and 40% of SHAP magnitude), consistent with their strong, monotone descriptive risk-response curves. Claude Sonnet 5 is the only model whose profile matches the human ordering outright, with the opponent's previous action largest at 51%, slightly above the human 48%; Gemini-3.5-Flash-Lite is the next closest (34%), though its progress gap remains more influential than in the human model. GPT-5.4-nano has no comparably dominant predictor: importance is spread across progress gap, risk, round, and its own previous action, with opponent history at only 7% and the lowest above-chance predictive fit in the table (AUC 0.56, balanced accuracy 0.53), so its behaviour is comparatively difficult to reconstruct from these five mechanical state variables rather than clearly governed by any one signal.

Claude Opus 5 shows why these shares must be read against the design that produced them. Its 43% opponent share reads as strong reciprocity and is not: on the neutral lane it plays UNSAFE on essentially every low-risk decision and essentially none at high risk, so the risk treatment alone separates its choices, and since

Table 7: Population-specific random-forest prediction of UNSAFE play. Feature columns report shares of total mean absolute SHAP magnitude (rows sum to 100%, up to rounding); AUC and balanced accuracy are evaluated on the population-specific test split.

Population	Own previous	Opponent previous	Progress gap	Risk	Round	Test AUC	Balanced accuracy
Human	6%	**48%**	14%	12%	20%	0.62	0.54
GPT-5-nano	2%	4%	**44%**	16%	33%	0.82	0.50
GPT-5.4-nano	18%	7%	**33%**	22%	20%	0.56	0.53
Gemini-3-Flash	5%	15%	23%	**35%**	23%	0.95	0.81
Gemini-3.1-Flash-Lite	16%	7%	13%	**40%**	24%	0.94	0.81
Gemini-3.5-Flash-Lite	14%	**34%**	27%	17%	9%	0.80	0.70
Claude Opus 5	26%	43%†	0%	30%	0%	1.00	1.00
Claude Sonnet 5	22%	**51%**	4%	14%	10%	0.96	0.92

†Collinear with the risk treatment, not a reciprocity estimate; see the discussion below.

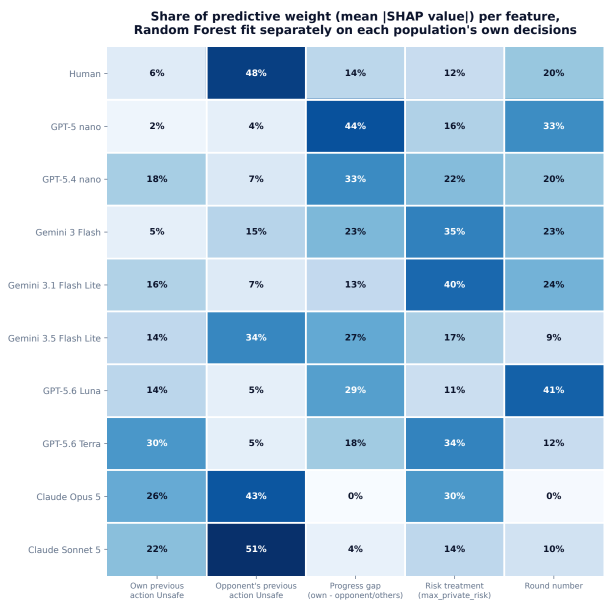

Figure 4: Population-specific predictive structure for round-$t \geq 2$ UNSAFE choices. Each row reports the share of mean absolute SHAP value assigned to the same five pre-decision features by a random forest fitted separately to that population. SHAP magnitudes describe how the fitted classifier uses observed features; they are neither causal effects nor evidence of the agents' internal reasoning.

both seats behave that way the opponent's previous action is a proxy for the risk level rather than an independent signal. The forest then splits importance between two variables encoding the same event, and the perfect AUC describes a step function, not a better fit. A feature-importance share is interpretable only where the feature keeps variation independent of the others, which is what a near-deterministic policy removes.

Predictive performance varies substantially across populations (AUC 0.56 to 1.00), so SHAP shares should not be read as directly comparable measures of decision quality; floor and ceiling behaviour mechanically limits minority-class prediction for several models. The defensible conclusion is narrower than "LLMs use different features": the same observable state variables organise predictive information differently across populations, and matching a human aggregate action rate does not imply matching the human reciprocity-dominated association profile.

A.8 Two-player checkpoint descriptive baseline (full scope)

The five-checkpoint two-player descriptive baseline referenced throughout the two-player strategic diversity results in the main paper and the observable drivers of unsafe play in the main paper completed 150 races and 2,790 decisions, crossing five checkpoints (GPT-5-nano, GPT-5.4-nano, Gemini-3-Flash-preview, Gemini-3.1-Flash-Lite, Gemini-3.5-Flash-Lite) with the mechanism's three assigned private-risk levels (0.10, 0.60, 0.90; Eq. (2)). At the level of descriptive curves rather than the pooled aggregate rates already reported in the main text, GPT-5-nano's UNSAFE rate was low and nearly flat across the three risk levels; GPT-5.4-nano's response to risk level was non-monotone; and the three Gemini checkpoints' UNSAFE rate declined as the assigned maximum risk increased – the opposite direction from the two OpenAI checkpoints. Because provider routes, protocol signatures, and pilot sample sizes differ across checkpoints, these five curves demonstrate qualitative heterogeneity in how each checkpoint responds to the risk manipulation; we do not pool them into a single model-family estimate, and the formal test of that heterogeneity is the checkpoint-by-risk interaction reported in the two-player strategic diversity results in the main paper ($\chi^2(10) = 354.7$). Table 8 gives the per-checkpoint, per-risk-level rates underlying the qualitative curve descriptions above.

Table 8: Player-level UNSAFE rate by checkpoint and assigned maximum-private-risk level, five-checkpoint two-player descriptive baseline ($n = 20$ players per checkpoint–risk cell, ten races each).

Checkpoint	Risk 0.10	Risk 0.60	Risk 0.90
GPT-5 nano	12.3%	15.1%	14.5%
GPT-5.4 nano	57.7%	49.6%	56.7%
Gemini 3 Flash	100.0%	72.3%	53.9%
Gemini 3.1 Flash Lite	100.0%	80.1%	69.9%
Gemini 3.5 Flash Lite	83.8%	70.8%	62.6%

A.9 N-player self-play scope ($N = 3, 4, 5$)

the multi-agent position results in the main paper reports position effects for GPT-5-nano and GPT-5.4-nano at fixed N; this subsection gives the broader group-size scope referenced there. Matched OpenAI self-play baselines cover $N \in \{3, 4, 5\}$ for both checkpoints: 180 pilot races, 720 player trajectories, and 6,120 decisions in total, pooled across the mechanism's three risk levels. Table 9 reports the aggregate UNSAFE rate by checkpoint and group size.

Table 9: Aggregate UNSAFE rate by group size, OpenAI self-play, pooled across the three assigned private-risk levels. Ten independent races per model–risk–N cell.

Checkpoint	$N = 3$	$N = 4$	$N = 5$
GPT-5-nano	16.2%	26.8%	21.9%
GPT-5.4-nano	83.7%	87.0%	84.0%

Neither checkpoint's response to group size is monotone in N, and the association is checkpoint-specific: GPT-5-nano's rate roughly doubles from $N = 3$ to $N = 4$ before partially receding at $N = 5$, while GPT-5.4-nano stays within a narrow 83–87% band throughout. We do not read either pattern as an isolated group-size effect, for two reasons. First, changing N also changes the stage-payoff rule itself (Eq. (3)–(4) use the joint count k^t of Safe-choosing companies rather than a single opponent's move), the number of opponents, and prompt length, so N is confounded with the representation of the game as well as with its size. Second, the pilots use only ten independent races per model–risk–N cell, which is a small base for the per-cell rates entering Table 9 and leaves individual cells vulnerable to the same kind of small-n noise flagged for specific rank × risk-band cells in the multi-agent position results in the main paper. Table 10 disaggregates Table 9 by assigned risk level and reports a 95% Wilson interval for every cell.

Table 10: UNSAFE rate by checkpoint, group size, and assigned maximum-private-risk level, OpenAI N-player self-play (95% Wilson intervals; n is the decision count per cell).

N	Checkpoint	Risk	n	UNSAFE rate (95% CI)
3	GPT-5 nano	0.10	255	17.3% (13.1–22.4%)
3	GPT-5 nano	0.60	255	18.0% (13.8–23.2%)
3	GPT-5 nano	0.90	255	13.3% (9.7–18.1%)
3	GPT-5.4 nano	0.10	255	84.3% (79.3–88.3%)
3	GPT-5.4 nano	0.60	255	84.3% (79.3–88.3%)
3	GPT-5.4 nano	0.90	255	82.4% (77.2–86.5%)
4	GPT-5 nano	0.10	340	21.2% (17.2–25.8%)
4	GPT-5 nano	0.60	340	30.3% (25.7–35.4%)
4	GPT-5 nano	0.90	340	28.8% (24.3–33.9%)
4	GPT-5.4 nano	0.10	340	87.6% (83.7–90.7%)
4	GPT-5.4 nano	0.60	340	88.8% (85.0–91.7%)
4	GPT-5.4 nano	0.90	340	84.4% (80.2–87.9%)
5	GPT-5 nano	0.10	425	20.2% (16.7–24.3%)
5	GPT-5 nano	0.60	425	21.2% (17.6–25.3%)
5	GPT-5 nano	0.90	425	24.2% (20.4–28.5%)
5	GPT-5.4 nano	0.10	425	80.2% (76.2–83.7%)
5	GPT-5.4 nano	0.60	425	87.1% (83.5–89.9%)
5	GPT-5.4 nano	0.90	425	84.7% (81.0–87.8%)

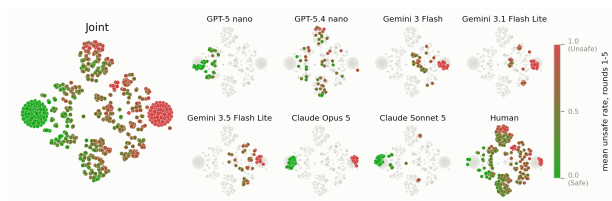

Figure 5: The same t-SNE coordinates as the population-identity embedding in the main paper, recoloured by each player's own mean UNSAFE rate over rounds 1–5 (green = SAFE, red = UNSAFE) instead of population identity. Left: all 760 trajectories pooled. Right: the same colouring broken out per population.

A.10 Trajectory embedding recoloured by outcome

The main paper reports that the embedding of raw five-round trajectories separates almost entirely by how much UNSAFE play a trajectory carries. This is that check: the same coordinates, with only the colouring changed, so any structure visible here is structure the embedding already had.

A.11 Mean round-by-round profile per population

The embedding in the main paper shows that human trajectories spread more widely than any single model's. This is the same comparison read one feature at a time: each population's mean profile against the pooled mean over the fourteen features every trajectory shares.

A.12 Unsafe rate by persona band and relative position

The main paper plots these rates as grouped bars for both checkpoints and reports the pattern they show. The tables below give the cell-level numbers behind that figure, so a single rate, its sample size and its confidence interval can be checked directly.

A.13 Behavioural-archetype coverage by population

The main paper reports that human trajectories spread across more archetypes than any single checkpoint's do. This table is the per-population breakdown behind that statement, so a single share can be checked against its population and archetype.

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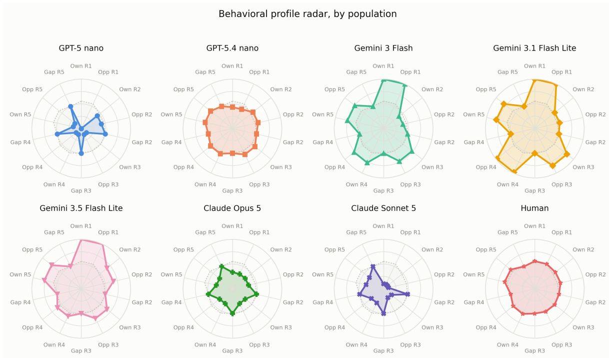

Figure 6: Mean round-by-round profile for each population, on the 14 features shared across every trajectory (own and opponent action at rounds 1–5, plus progress gap entering rounds 2–5; the round-1 gap is omitted because every race starts at gap 0 by construction and is therefore uninformative). Solid lines show the population mean; the dotted line is the pooled mean across all eight populations.

Table 11: UNSAFE rate by persona band and relative position, GPT-5-nano. n is the number of decisions in the cell; CI is the 95% confidence interval on the unsafe rate.

Persona band	Position	n	UNSAFE rate	95% CI
Baseline	Leader	1460	19.7%	17.7–21.8%
Baseline	Middle	1065	30.4%	27.7–33.3%
Baseline	Trailer	175	33.1%	26.6–40.4%
Low (R1–R2)	Leader	1094	5.9%	4.6–7.4%
Low (R1–R2)	Middle	198	14.6%	10.4–20.2%
Low (R1–R2)	Trailer	58	12.1%	6.0–22.9%
Mid (R3–R4)	Leader	649	40.7%	37.0–44.5%
Mid (R3–R4)	Middle	451	38.1%	33.8–42.7%
Mid (R3–R4)	Trailer	250	32.8%	27.3–38.8%
High (R5–R6)	Leader	702	63.2%	59.6–66.7%
High (R5–R6)	Middle	339	54.3%	49.0–59.5%
High (R5–R6)	Trailer	309	50.2%	44.6–55.7%

Table 12: UNSAFE rate by persona band and relative position, GPT-5.4-nano. Cells marked † have $n < 20$.

Persona band	Position	n	UNSAFE rate	95% CI
Baseline	Leader	1649	83.6%	81.8–85.3%
Baseline	Middle	703	84.8%	81.9–87.2%
Baseline	Trailer	348	86.5%	82.5–89.7%
Low (R1–R2)	Leader	1073	3.3%	2.4–4.5%
Low (R1–R2)	Middle	268	1.5%	0.6–3.8%
Low (R1–R2)	Trailer	9†	0.0%	0.0–29.9%
Mid (R3–R4)	Leader	1164	95.7%	94.4–96.7%
Mid (R3–R4)	Middle	70	88.6%	79.0–94.1%
Mid (R3–R4)	Trailer	116	91.4%	84.9–95.3%
High (R5–R6)	Leader	1328	99.3%	98.7–99.6%
High (R5–R6)	Middle	4†	100.0%	51.0–100.0%
High (R5–R6)	Trailer	18†	100.0%	82.4–100.0%

Table 13: Behavioural-archetype coverage: share of each population's neutral-lane player-races assigned to each human-fit archetype. Human row is the reference clustering (n=341); each LLM row projects that checkpoint's own no-persona player-races (n=60 to 120) into the same space. The sample sizes are summarized in the evidence-strata table in the main paper. This table draws on the full nine-checkpoint cross-model pilot and therefore includes GPT-5.6 Luna and Terra, which are not part of the seven-model HDBSCAN/t-SNE/decision-tree comparison earlier in this subsection.

Population	Cautious	Aggr./recip.	Catch-up	Persister
Human (reference)	39.0%	49.9%	5.6%	5.6%
GPT-5-nano	99.2%	0%	0%	0.8%
GPT-5.4-nano	51.7%	33.3%	3.3%	11.7%
Gemini-3-Flash	1.7%	78.3%	20.0%	0%
Gemini-3.1-Flash-Lite	0%	83.3%	16.7%	0%
Gemini-3.5-Flash-Lite	0%	76.7%	23.3%	0%
GPT-5.6 Luna	3.3%	72.5%	24.2%	0%
GPT-5.6 Terra	14.2%	59.2%	26.7%	0%
Claude Opus 5	66.7%	33.3%	0%	0%
Claude Sonnet 5	77.5%	20.8%	1.7%	0%

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